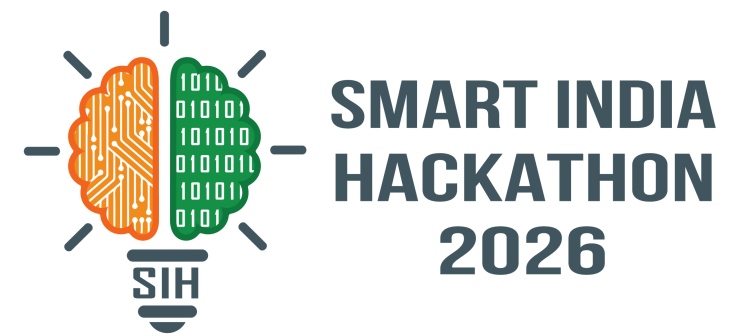




SMART INDIA HACKATHON 2026



- Problem Statement ID - **SIH26084**
- Problem Statement Title - **Convective scale nowcasting for Thunderstorms, Hail & Cloudbursts (06 hr)**
- Theme - **Disaster Management**
- PS Category - **Software**
- Team ID - **137710**
- Team Name - **TRIKAAL**

PROBLEM EXISTING

- **Too Coarse and Slow** : NWP resolves about **11–25 km**; which cannot resolve **1–10 km, 15–180 min** storms.
- **Scattered data** : Radar, INSAT, lightning and NWP sit in separate systems with no common grid.
- **Black-box warnings** : Forecasters can't see why a cell was flagged.
- **Alert gaps** : Warnings are not localised, not multilingual, and there is no reviewed path from model to public alert.

PROPOSED SOLUTION

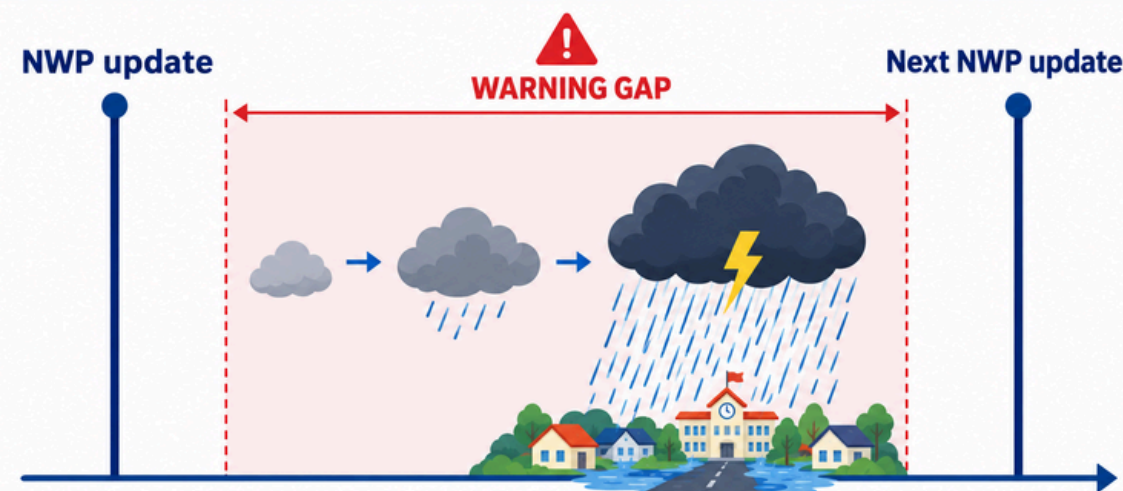
- **Dense optical flow** : Farneback tracks storm motion pixel by pixel from consecutive reflectivity grids.
- **Five hazard heads** : Hail, downburst, cloudburst, lightning density and convective initiation.
- **Explainable severity** : A **4-level** class (INFO/WATCH/WARNING/SEVERE) with real SHAP attributions.
- **Human-in-the-Loop CAP** : Officer approves, rejects or cancels, each with a reason and an audit trail.
- **1–3 km display grid** : Interpolated from NWP, and labelled as such.

WHAT MAKES VAYUGATI DIFFERENT

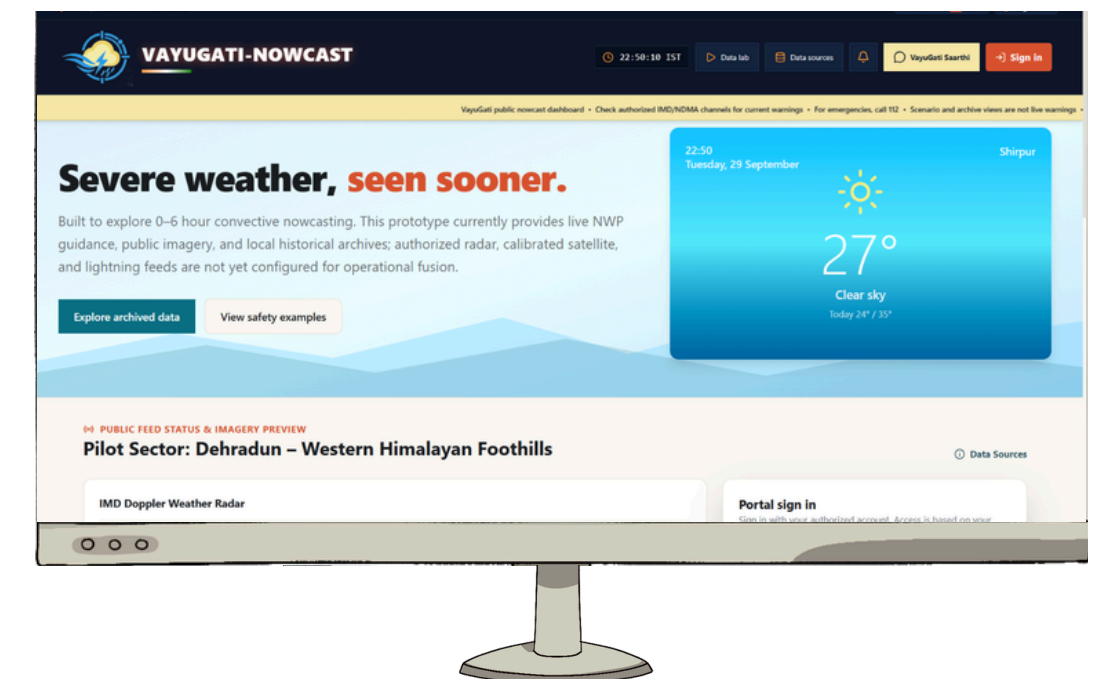
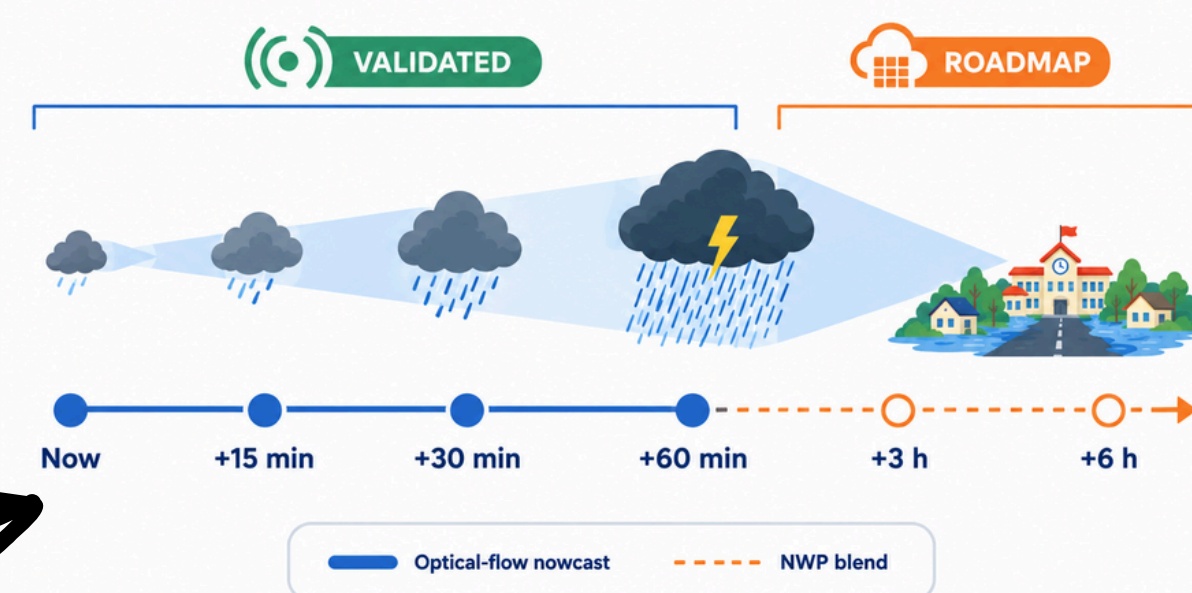
- **Proven, not claimed** : Scored against persistence (CSI, POD, FAR, FSS); test-storm speed recovered **within 1 km/h**.
- **Anchored to a real disaster** : Verified against the 2010 Leh cloudburst (**~200 dead**) using NASA GPM IMERG archives.
- **Governed** : The database enforces officer approval, with an immutable audit log.
- **Real multi-source data** : Live Open-Meteo NWP + archived MOSDAC INSAT-3DR + NASA IMERG, each status-labelled.
- **VayuGati Saarthi** : An AI chat guide that explains approved alerts and portal features in the user's language.

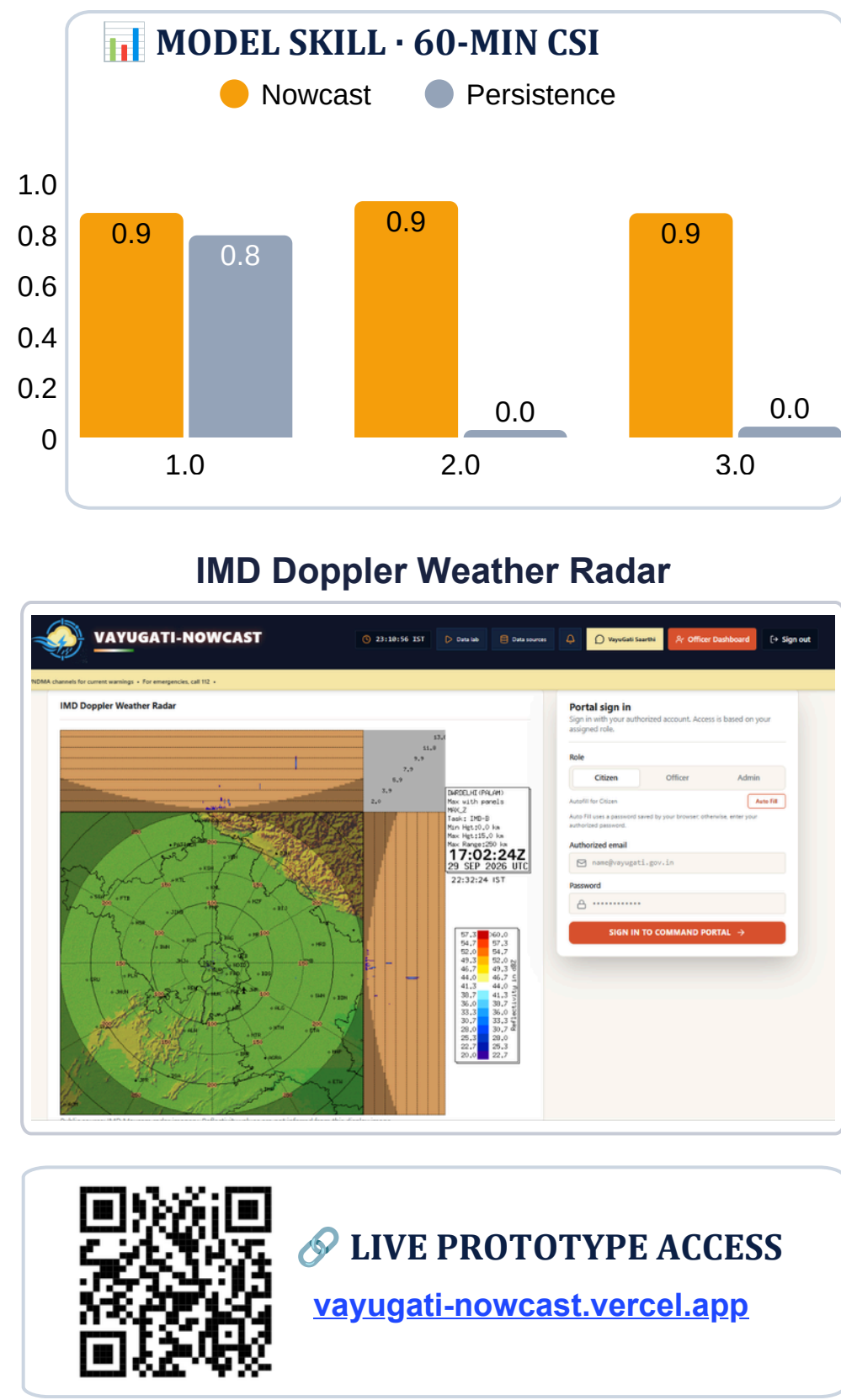
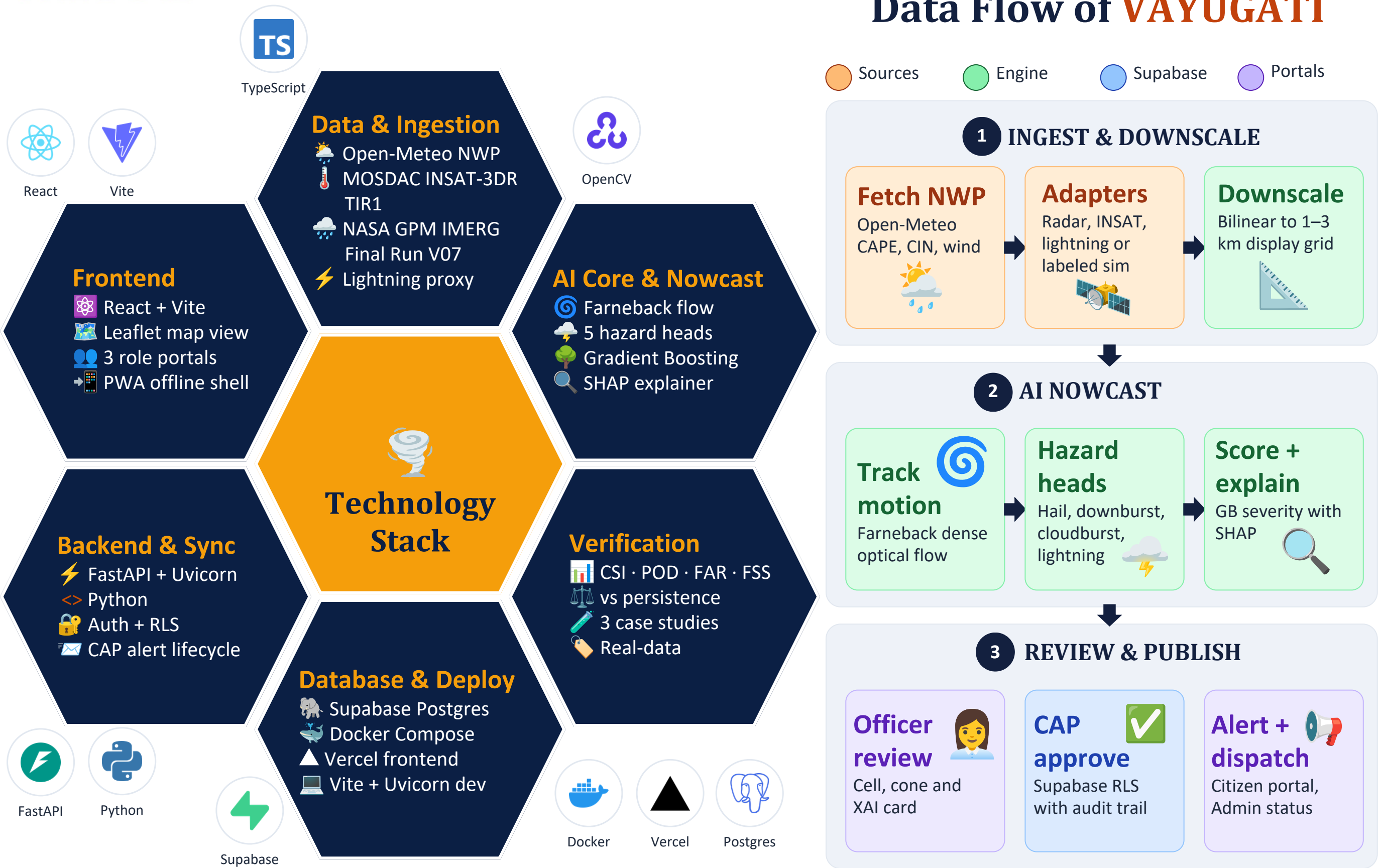


TODAY — THE PROBLEM



VAYUGATI — THE SOLUTION





FEASIBILITY AND VIABILITY

TECHNICAL

Open-source, CPU-only

- ✓ FastAPI + OpenCV Farneback + scikit-learn/SHAP run on CPU.
- ✓ Docker Compose build, adapters for IMD radar, INSAT and lightning feed one schema.

OPERATIONAL

Advisory overlay on today's process

- ✓ The forecaster reviews every alert; approve, reject or cancel with a reason.
- ✓ Role-based portals; no change to IMD field protocols.

ECONOMIC

Software only

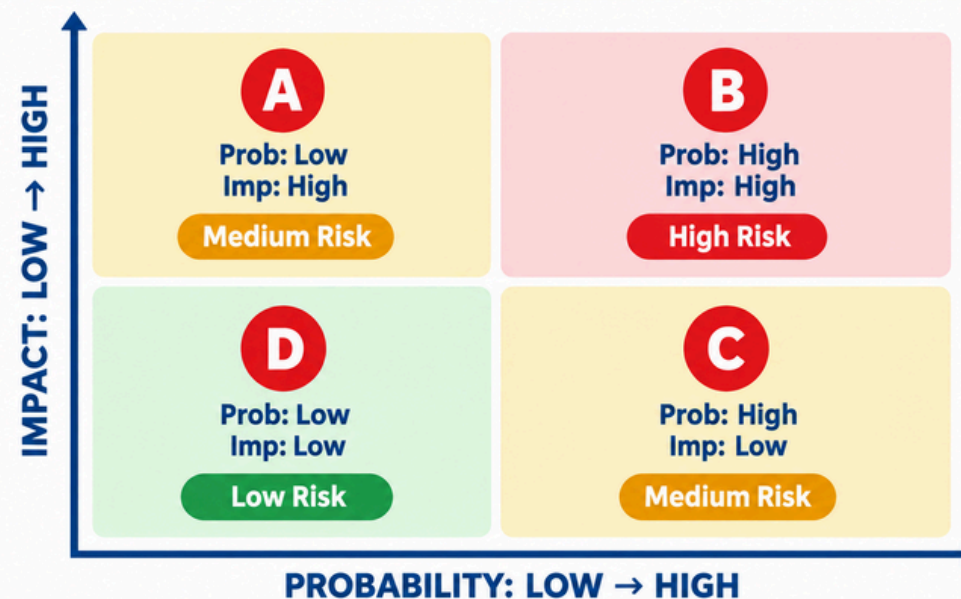
- ✓ No new radar, sensor or IoT hardware.
- ✓ Open-source stack; prototype runs on Supabase and Vercel tiers.

SAFETY

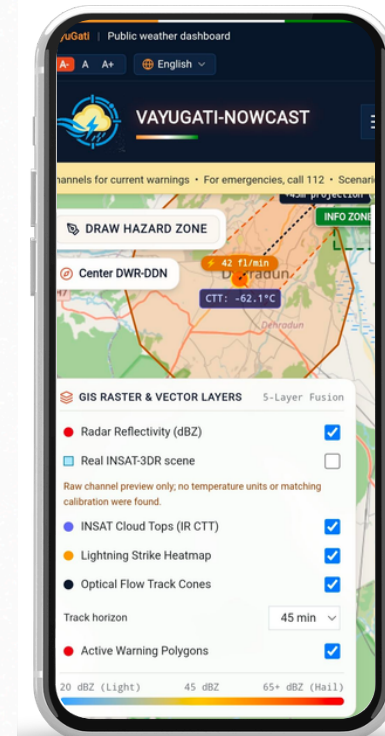
Human-gated, honestly labelled

- ✓ Database-enforced DRAFT → REVIEW → APPROVED, with an immutable audit log.
- ✓ LIVE / CACHED / OFFLINE / SIMULATED badge on every feed.

RISK MAP — CHALLENGES & MITIGATIONS (VAYUGATI NOWCAST)



LEGEND: A = Alert Error | B = Live Data Access | C = Model Quality | D = Adoption



DETAILED BREAKDOWN

- A** Wrong or missed warning could mislead the public
- B** Live IMD radar, satellite and lightning feeds need authorisation
- C** Model trained on synthetic labels; forecasts under-confident
- D** Forecasters may resist a new tool

MITIGATION / STRATEGY

- HUMAN GATE**
No alert without officer approval; reviewer, reason and time logged
- ADAPTER-READY**
Adapters built; read-only pilot after IMD approval; simulated data labelled
- DISCLOSED & ROADMAPED**
Verified vs persistence; next: recalibrate on real IMD events
- ADVISORY OVERLAY**
Officer keeps authority; SHAP reasons build trust



MEASURED PERFORMANCE

256 × 256
≈ 29 ms

512 × 512
≈ 108 ms

1024 × 1024
≈ 438 ms

1 CPU core

TRADITIONAL WEATHER WORKFLOW



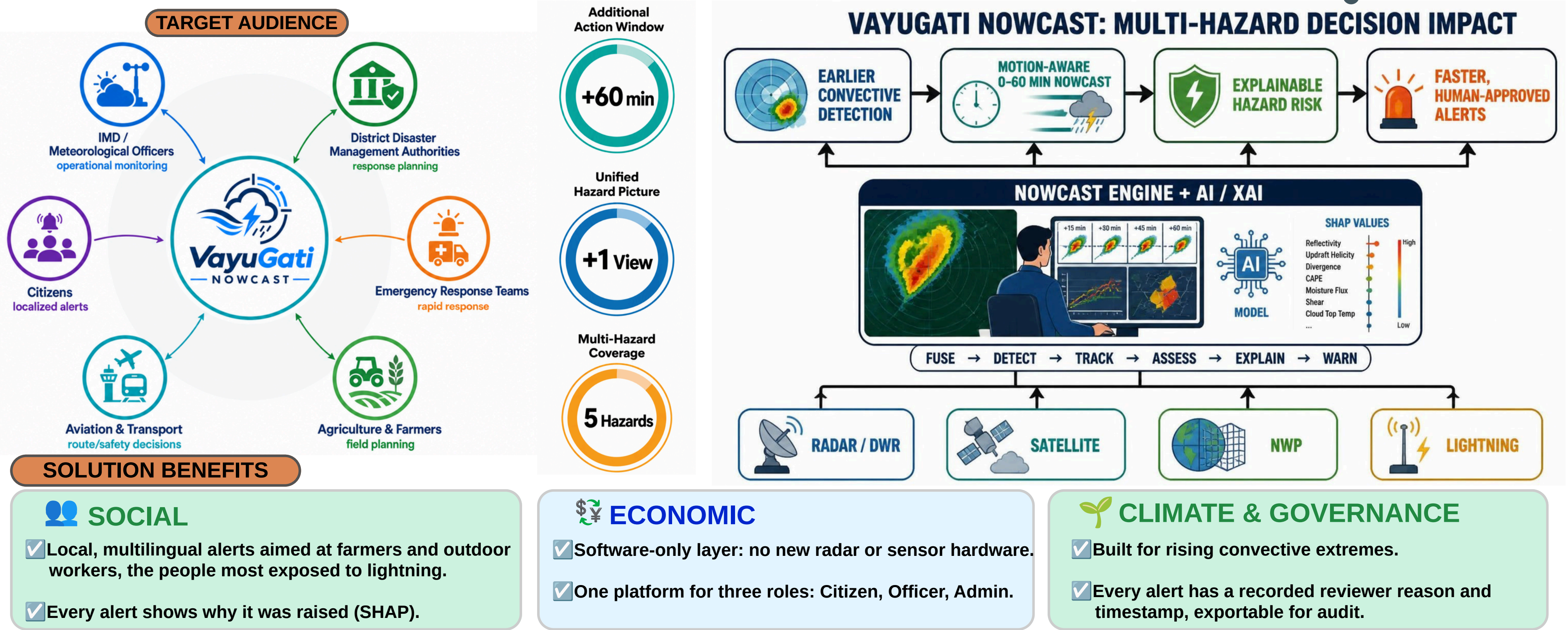
MULTIPLE WEATHER FEEDS

VAYUGATI NOWCAST



FUSION • ML • SHAP

 **VAYUGATI recommends; the Forecaster decides.** No alert reaches the public without approval.



STANDARDS & SAFETY BASIS



NDMA Lightning Guidelines (2020)

Lightning safety, public awareness and risk mitigation.



<https://ndma.gov.in/>



Common Alerting Protocol (CAP)

Standardized, interoperable emergency alert messages.



<https://docs.oasis-open.org/emergency/cap/v1.2/CAP-v1.2-os.html>



NCRB — Accidental Deaths & Suicides in India

Official statistics for lightning-related fatalities and disaster impact.



<https://www.ncrb.gov.in/>

REAL DATA SOURCES



NASA GPM IMERG — V07

Satellite-derived precipitation estimates for historical rainfall analysis and validation.



<https://gpm.nasa.gov/data/directory>



MOSDAC — INSAT-3D/3DR

Satellite imagery and meteorological products for cloud monitoring.



<https://mosdac.gov.in/>

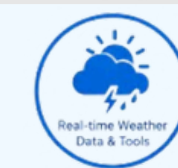


NASA GES DISC — Giovanni

Precipitation datasets, visualization and historical data analysis.



<https://giovanni.gsfc.nasa.gov/giovanni/>



Multi-Source Meteorological Data

Doppler Weather Radar • INSAT-3D/3DR • Lightning Detection Networks



<https://mausam.imd.gov.in/>

PROJECT RESOURCES



GitHub Repository

Source code, synthetic data, models & implementation



<https://github.com/mannbutani0306/vayugati-nowcast>



Project Report

Complete documentation of our problem, solution & development.



<https://drive.google.com/drive/folders/1LXmui8BvleWoOkqvVvavCsYsHd2yto1c?usp=sharing>



Prototype Demonstration

VayuGati Nowcast working prototype walkthrough video.



<https://youtu.be/LyKlljgVgBU>



Primary Case Study

Bhan, S.C., Devrani, A.K., Sinha, V. (2015). "An analysis of monthly rainfall and the meteorological conditions associated with cloudburst over the dry region of Leh (Ladakh), India." MAUSAM 66(1), 107–122.



<https://mausamjournal.imd.gov.in/index.php/MAUSAM/article/view/371/290>



Multi-source meteorological data fusion • AI-driven storm tracking • 0–6 hour nowcasting • Designed for 1–3 km hyper-local hazard mapping. Supporting safer, data-informed decisions through VAYUGATI-NOWCAST.



vayugati-nowcast.vercel.app/
Scan to try the live prototype
<https://vayugati-nowcast.vercel.app/>

Thank you: VAYUGATI-NOWCAST

Hyper-local Forecasts. Safer Tomorrows.

Team TRIKAAL • SIH26084 • Ministry of Earth Sciences (MoES) • Software